

Reweave: Negotiated-Congestion Rip-up-and-Reroute for Minor Embedding on D-Wave Quantum Annealers

ANONYMOUS AUTHOR(S)

Minor embedding—mapping a logical problem graph onto the fixed sparse connectivity of a quantum annealer by representing each logical variable as a connected *chain* of physical qubits—is the compilation bottleneck for quantum annealing, and *average chain length* (ACL) is the metric most widely taken as the primary proxy for embedded-problem solution quality. The de facto standard, *minorminer*, builds chains as unions of independent shortest paths under a memoryless per-pass congestion penalty; both choices inflate ACL and produce high run-to-run variance. We attack *minorminer* on two fronts—how it *builds* a chain and the *order* in which it searches—and, as a control, ask whether learning can beat it.

First, we observe that chain construction is exactly the multi-net global routing problem refined for thirty years in FPGA/VLSI computer-aided design, and port its two key ideas: a true Steiner-tree inner step in place of independent shortest paths, and *negotiated congestion* (McMurchie–Ebeling) in place of the snapshot penalty. We package these as **Reweave**, an anytime, large-neighbourhood *improver* that warm-starts from a fast base embedding and reroutes the longest chains through a congestion-priced fabric, committing a move only if it lowers qubit count and preserves validity—so it never returns an embedding worse than its base. On Erdős–Rényi, d -regular, and Barabási–Albert sources into clean and broken Pegasus and into Zephyr, reweave never lengthens ACL versus *minorminer* on any of 35 instance–topology cells; its thorough configuration lowers mean ACL on every cell (by 1–11%, mean $\sim 5\%$) and ACL standard deviation in 30 of 35 (by up to 77%), all at 100% success. A structured optimization pass—bounded routing, a dirty-set schedule, and spur-pruning—makes the production improver $\sim 3\times$ faster, within $\sim 1.3\times$ compiled *minorminer*’s wall-clock.

Second, we open *minorminer*’s *search order*—the lever its own authors named as the key open problem—with a ~ 90 -line fork that exposes the vertex order to its full search. A bandwidth-reducing (Cuthill–McKee) order, or a per-instance portfolio of such orders, lowers *minorminer*’s ACL by up to $\sim 5\%$ and roughly halves its run-to-run variance; it composes with Reweave and, unlike a learned or constructive embedder, never sacrifices *minorminer*’s reliability. On the *actual* hardware graphs (Advantage Pegasus-16, 5640 qubits; Advantage2 Zephyr-15, 7440 qubits) this guidance is essentially free: its wall-clock premium shrinks as problems grow, the ordered search running at or below *minorminer*’s time on the largest instances. (The complementary lever—which chain to rip up next—is, we find, inert.)

Finally, treating Reweave’s outputs as training data, we ask whether an *amortized* model can learn to embed, and report an honest, cautionary answer: the naive supervised approach is ill-posed—hardware placement is free under the fabric’s symmetries, so the model collapses to a constant—while a symmetry-invariant (Procrustes) objective repairs it and beats *minorminer* by $\sim 1\%$ at equal budget (statistically significant but small), and a decode-aware reinforcement-learning fine-tune does not exceed it. A strong heuristic with restarts thus remains hard to beat with learned placement. We then validate the premise that ACL is worth optimizing: on a simulator, across 2,736 embedded random-Ising instances, shorter-chain embeddings give significantly better solutions (Spearman $\rho = -0.63$ between ACL and ground-state probability, $p \approx 10^{-296}$, agreed by simulated annealing and spin-vector Monte Carlo), and we re-establish every empirical claim under a $K=15$ -instance redesign with confidence intervals and Holm-corrected paired significance tests. The algorithmic primitives throughout are classical; the contribution is their synthesis and empirical evaluation, inside the Ember benchmarking framework, for minor embedding.

CCS Concepts: • **Hardware** $\rightarrow$ **Quantum computation**; • **Theory of computation** $\rightarrow$ *Routing and layout*; • **Mathematics of computing** $\rightarrow$ *Graph algorithms*.

Additional Key Words and Phrases: minor embedding, quantum annealing, D-Wave, graph minors, negotiated congestion routing, Steiner tree, large-neighbourhood search

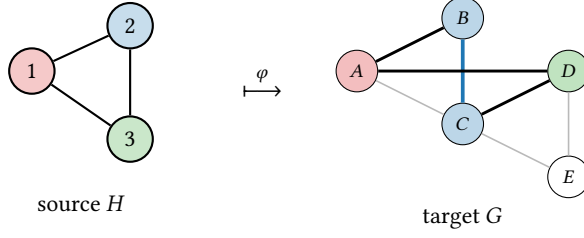


Fig. 1. Minor embedding of the triangle H into a hardware patch G . Vertex $1 \mapsto \varphi(1) = \{A\}$, $2 \mapsto \{B, C\}$, $3 \mapsto \{D\}$ (matching colours). Chain $\varphi(2)$ is connected through the bold intra-chain coupler $B-C$; the three chains are disjoint; each H -edge is realized by a hardware coupler between the corresponding chains (1–2 by $A-B$, 2–3 by $C-D$, 1–3 by $A-D$). Qubit E is unused. Here $ACL = \frac{1+2+1}{3} = \frac{4}{3}$.

1 INTRODUCTION

Quantum annealers such as those built by D-Wave solve optimization problems encoded as Ising/QUBO models whose interaction graph is the user’s *logical* (problem) graph. The hardware, however, exposes only a fixed, sparse *physical* graph—successive D-Wave generations Chimera, Pegasus [2], and Zephyr [3] have qubit degree 6, 15, and 20 respectively. Running a problem whose interaction graph is denser than the hardware requires *minor embedding*: each logical variable is realized by a connected set of physical qubits, a *chain*, whose qubits are forced into agreement by strong ferromagnetic couplings so that the chain behaves as one logical spin [6, 17].

The minor-embedding problem. Let H be the *source* (problem) graph and G the *target* (hardware) graph. A minor embedding is a map $\varphi : V(H) \rightarrow 2^{V(G)}$ assigning to each logical vertex v a chain $\varphi(v) \subseteq V(G)$, subject to three conditions:

- (1) **Connectivity.** Each $\varphi(v)$ induces a connected subgraph of G .
- (2) **Disjointness.** The chains are pairwise disjoint: $\varphi(u) \cap \varphi(v) = \emptyset$ for $u \neq v$.
- (3) **Edge preservation.** For every edge $(u, v) \in E(H)$ there is at least one hardware edge between the chains, i.e. some $a \in \varphi(u)$, $b \in \varphi(v)$ with $(a, b) \in E(G)$.

The map is total with non-empty chains, so every logical vertex is realized; Ember’s validator (§4) records this as the *coverage* check. Figure 1 illustrates the conditions. Feasibility is the question “does a minor embedding exist?”; in practice one wants the *best* embedding, because the dominant cost is qubit usage and chain fidelity. The standard objective is to minimize the *average chain length*

$$ACL(\varphi) = \frac{1}{|V(H)|} \sum_{v \in V(H)} |\varphi(v)|, \quad (1)$$

and to keep chain lengths *uniform* (low coefficient of variation): long and unequal chains require stronger chain couplings, shrink the available dynamic range for the logical problem, and empirically raise the annealer’s solution error. ACL is therefore the headline quality metric, with total qubit count, maximum chain length, and chain-length uniformity as close companions.

Contribution. This paper makes seven contributions, listed in the order they appear. (i) We frame minor-embedding chain construction as *multi-net global routing* and transfer two ideas from FPGA/VLSI computer-aided design—a real Steiner-tree inner step and *negotiated congestion*—to the quantum-annealing setting, packaging them as **Reweave**, an anytime, large-neighbourhood *improver* that warm-starts from a fast base embedding and, by construction, never regresses below it (§3). (ii) We evaluate Reweave inside the Ember benchmarking framework (§4) against minorminer

and minorminer-with-p-norm-layout on clean and broken Pegasus and on Zephyr (§5), showing it reduces mean ACL and—the property that most distinguishes it—run-to-run ACL *variance* in nearly every cell, while all methods retain full success and validity. (iii) We *optimize* the algorithm (§6): a structured pass yields three quality-safe improvements—bounded-region routing, a dirty-set incremental schedule, and a spur-pruning finishing pass—that together make the production algorithm $\sim 3\times$ faster at equal-or-better ACL, within $\sim 1.3\times$ compiled minorminer’s wall-clock. (iv) We ask whether *learning* helps (§7) and report an honest, cautionary answer: the naive supervised model is ill-posed and collapses, a symmetry-invariant (Procrustes) objective fixes it and beats minorminer by $\sim 1\%$ at equal budget (small but significant), decode-aware RL does not exceed it, and we retract an earlier unequal-budget comparison. (v) We open minorminer’s *search order* (§8) with a ~ 90 -line fork that injects a vertex order into its full search: a Cuthill–McKee order, or a per-instance portfolio, lowers its ACL by up to $\sim 5\%$ and roughly halves its variance, composes with Reweave (the cheap stack `reweave-mmfork-cuthill`), and stays as reliable as minorminer—while the complementary rip-up-order lever is inert. (vi) We study *scaling* on the actual hardware graphs (§9)—Advantage Pegasus-16 and Advantage2 Zephyr-15—and find the order-guided variants’ wall-clock premium *shrinks* with problem size, running at or below minorminer’s time on the largest instances. (vii) We *validate* the objective and harden the statistics (§10, §11): on a simulator, shorter-chain embeddings yield significantly better annealing solutions (Spearman $\rho = -0.63$ between ACL and ground-state probability, agreed by simulated annealing and spin-vector Monte Carlo), and a $K=15$ -instance redesign with confidence intervals and paired significance tests confirms the gains and the variance reduction—and shows which one variant does not survive the stronger design. Throughout we are explicit (§3.5) that the algorithmic primitives are not new; the novelty is their synthesis and application to minor embedding, together with the empirical findings.

2 BACKGROUND AND RELATED WORK

The standard heuristic and its weaknesses. minorminer (MM), the Cai–Macready–Roy heuristic [6], is randomized coordinate descent. It fixes a random vertex order and, for each vertex, rips out its chain and rebuilds it as the union of weighted shortest-path computations to its already-placed neighbours; qubit contention is discouraged by a vertex weight $\text{diam}(G)^k$ where k is the number of chains currently using the qubit, recomputed from scratch each pass; it iterates until the embedding is valid or a patience bound is hit. Three structural weaknesses follow directly. First, a chain is a *union of independent shortest paths*—a weak Steiner-tree heuristic that misses shared structure. Second, the congestion signal is *myopic*: a per-pass snapshot with no memory of how contested a qubit has been over time. Third, the result is highly *order- and seed-dependent*. The most recent state-of-the-art evaluation [8] documents the consequences on broken Pegasus: run-to-run ACL can differ by several qubits per chain on the same instance, MM is beaten by deterministic clique embedding once the source’s average degree exceeds the hardware degree, and MM routinely hits a ~ 1000 s cutoff on large dense instances. The MM paper itself names “better initial placement of vertex-models” as the key open problem.

Deterministic templates. For sufficiently structured inputs, embeddings can be read off a template. Clique (“busclique”) embedding generates complete-graph minors in Chimera and Pegasus connectivity directly [4]; complete-bipartite templates [26] and the 4-clique network on a contracted hardware graph [22] extend the idea. These are fast and uniform but over-provision qubits on graphs that are not near-complete.

Layout- and spring-seeded heuristics. A complementary line attacks MM’s initial-placement weakness by first laying out the source graph in a low-dimensional space and placing variables near the geometrically closest qubits: layout-aware embedding [23], the spring/clique-seeded methods

of Zbinden et al. [32], and Ocean’s “Layout Embedding”, which seeds MM with a p -norm layout. We use the latter as a baseline.

Exact methods. Integer-programming formulations with Benders decomposition [1] and algebraic-geometry/Gröbner-basis compilers [7] solve minor embedding optimally but do not scale past tens of vertices.

Metaheuristics and learned policies. Path/super-vertex simulated annealing (PSSA) [27] and graph-theoretic OCT/virtual-hardware methods [9] apply local search; more recently, GNN-RL (CHARME [21]) and PPO policies [20] learn chain-construction heuristics.

Routing and optimal transport. Two threads inform our approach. Negotiated-congestion routing (PathFinder) [18] from FPGA CAD treats each net as a resource consumer that negotiates for congested wires over iterations; and Gromov–Wasserstein optimal transport [30, 31] matches graphs by their intra-graph distances, suggesting a global, continuous view of the assignment. Our method develops the first thread; the second is promising future work (§12).

3 THE PRELIMINARY REWEAVE ALGORITHM

This section presents Reweave as first designed; §5 reports its results. A subsequent optimization pass (§6) makes the production algorithm $\sim 3\times$ faster at equal-or-better quality without changing the ideas below, so we keep this preliminary description intact and report the optimized numbers separately.

3.1 Minor embedding as multi-net routing

A chain $\phi(v)$ that must be connected and adjacent to each neighbour’s chain is exactly a *Steiner tree* (a routing “net”) whose terminals are the qubits bordering the neighbour chains, and the disjointness requirement is exactly the rule that two nets may not share a routing resource. Minor embedding is therefore a multi-net global routing problem on the graph G , and the mature machinery of FPGA/VLSI routing applies. We import its two central ideas.

3.2 A real Steiner inner step

Rather than build a chain as a union of independent shortest paths to a single root (MM), we grow one tree with the classical *shortest-path heuristic* (SPH) [19, 28]: starting from a root, each neighbour terminal is connected to the *nearest node of the current tree*, so later connections reuse earlier structure. Figure 2 contrasts the two: when neighbours cluster, SPH shares a trunk and produces a strictly shorter chain.

3.3 Negotiated congestion

We replace MM’s snapshot penalty with the negotiated-congestion cost of McMurchie and Ebeling [18]. Routing prices each qubit q at

$$\text{cost}(q) = (1 + h_q) (1 + \lambda \cdot o_q), \quad (2)$$

where o_q is the number of *other* chains currently occupying q (occupancy), λ is the present-congestion factor, and h_q is a *history* term that *accumulates* by a fixed increment every iteration in which q is over-subscribed. The present term makes a qubit more expensive as nets pile onto it within a pass; the history term gives the price a memory, so a persistently contested qubit becomes permanently expensive and the chains that can most cheaply route around it eventually yield. This is a Lagrangian view of the disjointness constraint; in FPGA routing it is known to drive the search toward a legal, low-congestion state, and we observe the same behaviour here—behaviour

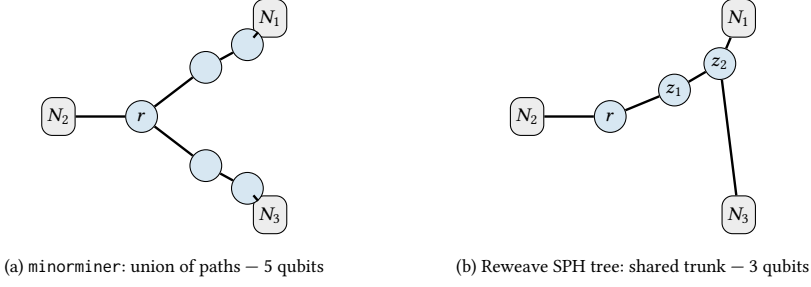


Fig. 2. Inner step contrast for a chain that must reach three neighbour chains N_1, N_2, N_3 . (a) MM routes an independent shortest path to each, duplicating qubits when N_1 and N_3 are near each other. (b) The shortest-path-heuristic Steiner tree attaches N_3 to the nearest existing tree node (z_2), reusing the trunk built for N_1 , yielding a shorter chain.

MM’s memoryless snapshot cannot sustain. A node-weighted Dijkstra over Eq. (2) supplies the shortest-path computations the SPH step needs.

3.4 A warm-started large-neighbourhood improver

A from-scratch router that places every chain from scattered single-qubit seeds is throttled by exactly the open problem MM names—*initial placement*. Without co-locating adjacent variables, dense-graph chains balloon and the search often fails to legalize at all. The standalone reweave-cold variant (a cold start followed by the same improver) bears this out: on the nine clean-Pegasus Erdős–Rényi instances of our grid ($n \in \{20, 30, 40\}$, $d \in \{0.3, 0.5, 0.7\}$) it legalizes only four—the smaller and sparser ones, at ACL 2.5–3.8—and fails on the five larger, denser ones, whereas minorminer embeds all nine (reproduced by docs/paper/data/coldstart_probe.py). Reweave therefore runs as an *improver*. It warm-starts from a fast base embedding (MM by default; any registered algorithm works), then performs *large-neighbourhood* rip-up-and-reroute—the neighbourhood being a long chain together with the cascade of chains its rerouting displaces. It takes the longest chain and lets it re-route along the cheapest tree under Eq. (2)—which may cross occupied territory and create temporary overlaps—then re-routes every chain it displaced through free space. A move is committed only if it lowers the embedding’s total qubit count (equivalently, its ACL) and leaves it valid; otherwise the prior state is restored (Algorithms 1–2). Because Reweave keeps the best valid embedding it has seen, it **never returns an embedding worse than its base**, and it spends its entire time budget reducing ACL—an anytime guarantee MM’s one-shot, patience-bounded run lacks. The thorough configuration additionally runs four seeded restarts and keeps the best, trading wall-clock for lower ACL and variance. When no valid base is available, a fallback builds the initial embedding from scratch in breadth-first vertex order under the same negotiated cost; uncompetitive on its own, it exists only to guarantee an output.

3.5 Novelty

We state plainly what is and is not new. Negotiated-congestion routing is due to McMurchie and Ebeling [18]; the shortest-path Steiner heuristic is classical [19, 28]; and large-neighbourhood/destroy-repair search is a standard matheuristic paradigm. *None of these primitives is our invention*. The contribution is their synthesis and their application to quantum-annealing minor embedding: casting chain construction as multi-net routing, replacing MM’s two weakest components (its inner step and its congestion signal) in a single drop-in algorithm, running it as a warm-started anytime improver with a never-regress guarantee, and validating empirically that this reduces

Algorithm 1 Reweave (driver)

Require: source H , target G , base embedder $\mathcal{B}$

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1:  $\Phi \leftarrow \mathcal{B}(H, G)$  ▷ warm start
2: if  $\Phi$  invalid then  $\Phi \leftarrow \text{COLDSTART}(H, G)$  ▷ BFS-ordered negotiated fallback
3: end if
4: if  $\Phi$  invalid then return failure
5: end if
6:  $\Phi^* \leftarrow \Phi$  ▷ best valid so far
7: repeat
8:   for all  $v \in V(H)$  in decreasing chain length do
9:      $\Phi \leftarrow \text{TRYSHORTEN}(\Phi, v)$  ▷ Alg. 2
10:    if  $\text{ACL}(\Phi) < \text{ACL}(\Phi^*)$  then  $\Phi^* \leftarrow \Phi$ 
11:    end if
12:  end for
13: until no improvement or deadline
14: return  $\Phi^*$ 

```

Algorithm 2 TRYSHORTEN(Φ, v) — one LNS destroy-repair move

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1: snapshot  $\Phi$ 
2: rip up  $\varphi(v)$ ; rebuild it as the cheapest SPH tree under Eq. (2) ▷ may overlap others
3: if new chain not shorter then restore; return  $\Phi$ 
4: end if
5: for all chain  $w$  now overlapping  $\varphi(v)$  do
6:   rip up  $\varphi(w)$ ; reroute it through free qubits only
7:   if no free-space route then restore; return  $\Phi$ 
8:   end if
9: end for
10: if total qubits decreased and  $\Phi$  valid then return  $\Phi$ 
11: else restore; return  $\Phi$ 
12: end if

```

ACL and—more importantly—ACL variance against the standard tool. The clean substitution also isolates a question of independent interest: *is it MM’s congestion signal or its inner step that costs the most?*

4 EMBER AND EVALUATION METHODOLOGY

We evaluate inside **Ember** (ember-qc), an open benchmarking framework for minor-embedding algorithms on D-Wave topologies. Every algorithm implements a common interface and is run through Ember’s atomic harness `benchmark_one`, which times the call and computes quality metrics (chain lengths, ACL, max chain length, qubits, couplers). Before any metric is trusted, the output passes four-layer validation: type and format checks followed by the structural checks of §1 (coverage, connectivity, disjointness, edge preservation). Targets are generated with `dwave-networkx` and may have qubits removed to simulate fabrication faults, and per-trial seeds are derived deterministically by SHA-256 of the task identity, so runs are reproducible and order-independent. Results aggregate into per-group means and standard deviations.

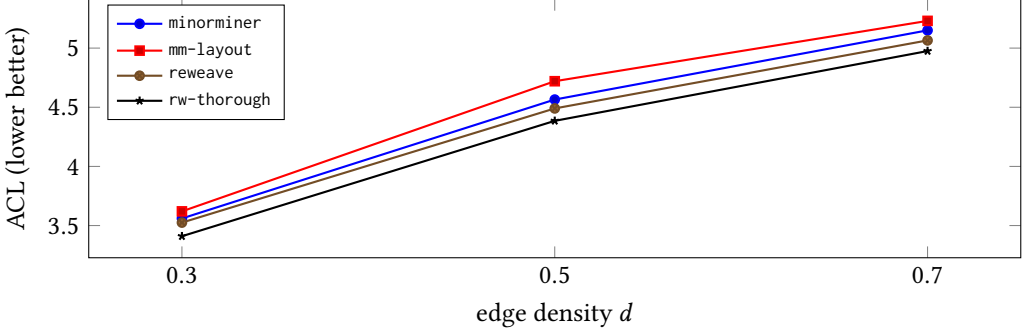


Fig. 3. ACL versus edge density for $n=40$ ER sources on clean Pegasus P_6 . reweave-thorough (bottom curve) is uniformly shortest; the gap to minorminer is roughly constant in absolute terms across densities.

Protocol. Targets are clean Pegasus P_6 , a *broken* Pegasus P_6 with 5% of qubits removed at random (the state-of-the-art evaluation setting), and Zephyr Z_4 . Sources are an Erdős-Rényi (ER) grid over $n \in \{20, 30, 40\}$ and edge density $d \in \{0.3, 0.5, 0.7\}$, plus d -regular and Barabási-Albert graphs for generality; one fixed instance per cell, so the reported spread is the run-to-run variance *on a fixed instance*—MM’s documented failure mode. Baselines are minorminer and minorminer-layout (p -norm layout). Each (algorithm, instance) is run over five seeds; we report success rate, ACL mean and standard deviation, max chain length, qubits, and wall-clock time.

5 PRELIMINARY RESULTS

These are the results of the preliminary algorithm of §3; the optimized production version and its (improved, and much faster) numbers follow in §6. All four methods embed every instance across all seeds and topologies (100% success), so success is not a differentiator on this grid; the comparison is on chain length, its variance, and time. Table 1 reports embedding quality on clean Pegasus P_6 , Table 2 the corresponding wall-clock time, and Table 3 robustness across topologies; Figure 3 shows the density trend.

Two findings are sharp. First, the single-seed reweave *never regresses*: across all 35 instance-topology cells (every per-seed run, in fact) its mean ACL is $\leq$ minorminer’s—the maximum observed difference is 0.000 to three decimals—confirming the best-valid guarantee empirically. Second, reweave-thorough lowers mean ACL on the six shown cells by 3.2–4.5%, and reduces ACL standard deviation in five of the six—by up to 76% on the full clean- P_6 grid (e.g. $0.11 \rightarrow 0.03$ at $n=30, d=0.3$). The lone clean- P_6 exception is $n=40, d=0.3$, where minorminer’s spread is already at the floor (0.05); we report it rather than hide it. The complete 35-cell grid across all three source families is in Appendix A: there reweave-thorough lowers mean ACL on *every* cell (by 1–11%; the largest single gain, -10.6% , is on a d -regular source) and cuts ACL standard deviation in 30 of the 35.

Discussion. The single-seed reweave is the fair *equal-base* comparison: it consumes the same minorminer base for a given seed and only refines it, so it cannot regress—and it already trims ACL on denser cells while leaving the sparse and d -regular cells (where the base is near-optimal) unchanged. The substantive gains come from reweave-thorough, whose four restarts both lower the mean and, by selecting the best of several runs, sharply cut the run-to-run variance that is minorminer’s signature weakness. The effect persists on broken Pegasus—where minorminer’s ACL inflates most and the thorough gain is largest—and on Zephyr, so the mechanism is not topology-specific. The cost is wall-clock time: Reweave layers a pure-Python router on top of the base, and

Table 1. Embedding quality on clean Pegasus P_6 for ER sources (mean over 5 seeds; ACL std in parentheses; lower ACL is better). Best ACL per cell in bold.

cell	algorithm	ACL (std)	max	qubits
$n=30, d=0.3$	minorminer	2.70 (0.11)	3.8	81.0
	minorminer-layout	2.75 (0.15)	4.0	82.4
	reweave	2.69 (0.10)	3.8	80.6
	reweave-thorough	2.59 (0.03)	3.8	77.6
$n=30, d=0.5$	minorminer	3.30 (0.16)	4.6	99.0
	minorminer-layout	3.43 (0.10)	4.8	102.8
	reweave	3.25 (0.15)	4.6	97.6
	reweave-thorough	3.19 (0.10)	4.0	95.8
$n=30, d=0.7$	minorminer	3.83 (0.22)	5.0	114.8
	minorminer-layout	3.80 (0.17)	5.4	114.0
	reweave	3.79 (0.23)	5.0	113.8
	reweave-thorough	3.65 (0.06)	4.8	109.6
$n=40, d=0.3$	minorminer	3.56 (0.05)	5.2	142.4
	minorminer-layout	3.62 (0.14)	5.4	144.8
	reweave	3.53 (0.06)	5.2	141.0
	reweave-thorough	3.41 (0.06)	5.0	136.4
$n=40, d=0.5$	minorminer	4.57 (0.10)	6.4	182.6
	minorminer-layout	4.72 (0.13)	6.8	188.8
	reweave	4.49 (0.14)	6.2	179.6
	reweave-thorough	4.39 (0.08)	6.2	175.4
$n=40, d=0.7$	minorminer	5.15 (0.06)	7.2	206.0
	minorminer-layout	5.23 (0.17)	7.0	209.2
	reweave	5.07 (0.04)	7.0	202.6
	reweave-thorough	4.98 (0.04)	7.0	199.0

Table 2. Wall-clock time (mean seconds per embedding, clean P_6). Reweave is pure Python; minorminer is compiled C++. reweave-thorough runs four seeded restarts and may use up to the full 20 s time budget.

cell	mm	mm-layout	rw	rw-thor.
$n=30, d=0.3$	0.33	0.33	1.11	20.08
$n=30, d=0.5$	4.44	3.30	9.04	18.82
$n=30, d=0.7$	1.22	0.98	3.23	12.33
$n=40, d=0.3$	0.54	0.46	2.12	8.37
$n=40, d=0.5$	1.11	0.99	4.12	16.41
$n=40, d=0.7$	1.35	1.73	5.87	20.01

reweave-thorough, with its four restarts, can use the full 20 s budget; minorminer-layout, by contrast, is neither consistently faster nor consistently better than plain minorminer.

6 OPTIMIZATIONS

The preliminary algorithm wins on chain length and variance but pays in wall-clock: its pure-Python router is $\sim 3\text{--}6\times$ slower than compiled minorminer (Table 2). We therefore ran a structured

Table 3. Robustness across topologies for the $n=40, d=0.5$ ER source: ACL (std) and time. minorminer’s ACL inflates most on broken Pegasus, where reweave-thorough’s relative gain is largest (-7.3%).

target	algorithm	ACL (std)	time (s)
Pegasus P_6	minorminer	4.57 (0.10)	1.11
	minorminer-layout	4.72 (0.13)	0.99
	reweave	4.49 (0.14)	4.12
	reweave-thorough	4.39 (0.08)	16.41
broken P_6 (5%)	minorminer	4.88 (0.18)	0.82
	minorminer-layout	4.72 (0.19)	1.19
	reweave	4.80 (0.18)	3.52
	reweave-thorough	4.52 (0.06)	14.26
Zephyr Z_4	minorminer	3.63 (0.08)	0.98
	minorminer-layout	3.61 (0.16)	1.10
	reweave	3.57 (0.07)	3.81
	reweave-thorough	3.52 (0.03)	13.90

optimization pass—eight candidate improvements (five targeting speed, three quality), each implemented as an isolated variant and measured against the frozen preliminary algorithm. Three quality-safe winners survived and are baked into the production algorithm; the rest were rejected or deferred.

What we bake in. The three production optimizations are mutually orthogonal (each rewrites a different stage of the router) and compose without interaction:

- **Bounded-region routing.** Each per-neighbour shortest-path search is restricted to a breadth-first ball around the vertex’s current chain and its placed-neighbour chains, and stops as soon as the neighbour boundary is settled, instead of computing a full single-source shortest-path tree over the whole 680–5640-qubit fabric. Alone this is $\sim 3\times$ faster with byte-identical chains.
- **Dirty-set incremental scheduling.** The improvement loop keeps a worklist of vertices whose neighbourhood changed rather than re-sweeping every chain each round; it performs the same accepted moves with fewer wasted re-examinations.
- **Spur-pruning.** A deterministic finishing pass removes redundant chain leaves—qubits whose removal keeps the chain connected and breaks no incident edge—to a fixpoint. This is the cheap shadow of exact region repair: on minorminer’s own output it recovers about half of a CP-SAT repair’s ACL gain at $\sim 5000\times$ less time, and on Reweave it shaves a further $\sim 0.6\%$ ACL for free.

A fourth idea—seeding the improver from a multilevel placement instead of minorminer—lowers ACL and variance on large, dense Pegasus instances (e.g. -6% ACL and -17% seed std at $n=40, d=0.7$ under the $K=15$ design of §11), but the multilevel base is unreliable on small, sparse, and Zephyr instances (it inflated ACL by up to 30% where minorminer is already near-optimal, so that pooled over the grid it is a net regression), so we keep it as an optional variant rather than the default. We also rejected an A^* routing scheme (a $+1.7\%$ ACL regression) and an exact Dreyfus–Wagner Steiner inner step (no gain over the shortest-path heuristic). A compiled (numba) node-weighted Dijkstra core, which we built and tested, gives *no* net speedup once bounded-region routing has localized each search (§12)—so the production router stays pure Python; only parallelized restarts remain as future speed work.

Table 4. **Optimized** Reweave, embedding quality on clean Pegasus P_6 (ER sources, mean over 5 seeds; ACL std in parentheses; lower is better). Best ACL per cell in bold. Cf. the preliminary Table 1; the optimized reweave matches-or-beats it everywhere at $\sim 3\times$ less time (Table 5).

cell	algorithm	ACL (std)	qubits
$n=30, d=0.3$	minorminer	2.70 (0.11)	81.0
	minorminer-layout	2.75 (0.15)	82.4
	reweave	2.68 (0.11)	80.4
	reweave-thorough	2.58 (0.03)	77.4
$n=30, d=0.5$	minorminer	3.30 (0.16)	99.0
	minorminer-layout	3.43 (0.10)	102.8
	reweave	3.23 (0.15)	97.0
	reweave-thorough	3.17 (0.11)	95.0
$n=30, d=0.7$	minorminer	3.83 (0.22)	114.8
	minorminer-layout	3.80 (0.17)	114.0
	reweave	3.77 (0.20)	113.0
	reweave-thorough	3.65 (0.07)	109.4
$n=40, d=0.3$	minorminer	3.56 (0.05)	142.4
	minorminer-layout	3.62 (0.14)	144.8
	reweave	3.51 (0.06)	140.4
	reweave-thorough	3.39 (0.06)	135.6
$n=40, d=0.5$	minorminer	4.57 (0.10)	182.6
	minorminer-layout	4.72 (0.13)	188.8
	reweave	4.45 (0.11)	177.8
	reweave-thorough	4.35 (0.08)	174.0
$n=40, d=0.7$	minorminer	5.15 (0.06)	206.0
	minorminer-layout	5.23 (0.17)	209.2
	reweave	4.95 (0.06)	198.0
	reweave-thorough	4.87 (0.03)	194.8

Optimized results. Tables 4 and 5 report the optimized algorithm on the clean-Pegasus grid. The optimizations are quality-safe: optimized reweave is never worse than the preliminary engine and, thanks to spur-pruning, slightly better—mean ACL -1.3% vs. minorminer over all 35 cells (vs. -0.9% for the preliminary engine), and -2.1% vs. minorminer-layout. The headline is speed: single-seed reweave now runs in $\sim 1.3\times$ minorminer’s wall-clock (down from $\sim 3\text{--}5\times$) while producing shorter chains, and the optimized reweave-thorough delivers -5.1% mean ACL vs. minorminer (up to -10.6% ; -5.9% vs. layout) at roughly the former cost of a *single* preliminary run rather than four. The optimized Reweave thus keeps the quality advantage of §5 while closing most of the wall-clock gap to compiled minorminer.

7 A LEARNING-BASED APPROACH

Reweave is a hand-engineered improver. A complementary question is whether an *amortized* model can *learn* to embed—paying an offline training cost once and then predicting an embedding in a single forward pass, rather than searching at solve time. This amortized premise—a deep network that accelerates graph embedding—is precisely that of a recent patent application [11]. The cache of problems Reweave has already solved is a ready-made training set—and reusing embeddings cached from previously-solved problems is itself a patented strategy [5]: each (*problem graph*,

Table 5. Wall-clock after optimization (mean seconds per embedding, clean P_6). “prelim.” is the preliminary single-seed engine (reweave-base). Optimized reweave matches minorminer to within $\sim 1.3\times$ and is $\sim 3\times$ faster than the preliminary engine; optimized reweave-thorough (best-of-4) costs about what a single preliminary run did.

cell	mm	prelim.	rw	rw-thor.
$n=30, d=0.3$	0.22	0.94	0.27	1.08
$n=30, d=0.5$	0.70	2.08	0.75	2.31
$n=30, d=0.7$	0.83	2.82	1.03	3.41
$n=40, d=0.3$	0.53	2.12	0.68	2.40
$n=40, d=0.5$	1.15	3.86	1.33	5.08
$n=40, d=0.7$	1.33	5.59	1.87	8.41

Reweave-optimized embedding) pair is a labelled example, and we ask a model to reproduce—and ideally improve on—that quality. This mirrors the predict-then-refine pattern of learned embedding heuristics [20, 21], where a network proposes a candidate that a classical minor-embedding step then refines.

We implemented several model families *inside Ember*, so they are benchmarked through the exact same harness (§5) as minorminer (MM) and Reweave. The honest finding is a cautionary one with a useful twist. The obvious supervised approach—regress each vertex’s hardware coordinate from a graph-VAE or GNN—is **ill-posed and collapses to a constant** (§7.6), and an early “the VAE beats Reweave-thorough” reading was an artifact of an unequal restart budget, which we retract. The fix is a **symmetry-invariant (Procrustes) objective** (§7.6) that recovers a real layout and beats minorminer by $\sim 1\%$ at equal budget—statistically significant but small, and a decode-aware reinforcement-learning fine-tune does not exceed it. We report the whole arc, because the failure mode is as instructive as the fix.

7.1 Embedding as a learned layout

The central reduction is to predict a *layout* rather than an embedding directly. Every family maps each logical vertex v to a coordinate $\hat{p}(v) \in [0, 1]^2$ in the target’s normalized hardware frame (read off the Pegasus/Zephyr drawing layout). Two decoders turn a layout into a valid embedding, both reusing Reweave’s repair backend (§3) so that no validity logic is re-implemented: *seed*→*MM* snaps each vertex to a distinct nearest qubit (an optimal linear-assignment problem) and hands those single-qubit chains to minorminer as *initial_chains*; *direct*→*repair* forms a proximity soft-assignment and rounds, grows, and legalizes it. Feeding the *true* Reweave chain centroids through the seed decoder recovers Reweave-quality embeddings (mean ACL within $1.03\times$, 100% valid), so the decoder is near-lossless: the learning problem is exactly “predict a good layout.”

7.2 Background: variational autoencoders on graphs

A variational autoencoder (VAE) [14] learns a probabilistic encoder $q_\phi(z | x)$ and decoder $p_\theta(x | z)$ over a latent variable z , trained by maximizing the evidence lower bound

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)] - \beta \text{KL}(q_\phi(z | x) \| p(z)),$$

a reconstruction term traded against a standard-normal prior $p(z)$. Graph VAEs replace the encoder with a graph neural network and have been used to embed and generate graphs [15, 25]. Our encoder is a message-passing GNN (a residual GraphSAGE stack [10]; a graph-attention encoder [29] is a drop-in alternative). The twist is twofold: the decoder emits a hardware *layout* rather than a reconstructed graph, and we exploit the model’s *generative* nature at inference time—sampling

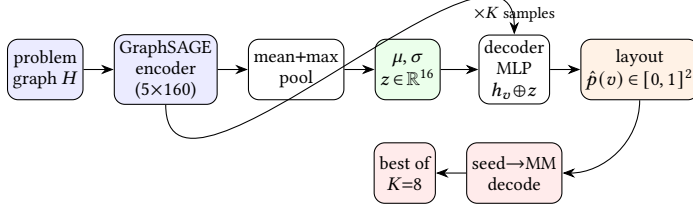


Fig. 4. learned-vae. A graph-VAE encodes H to a latent z ; the decoder turns z (plus per-node features) into a hardware *layout*; $K=8$ sampled layouts are each decoded to a valid embedding via seed→minorminer, and the shortest-ACL one is returned.

several latents yields several candidate layouts, and we keep the best after decoding. This is a cheap, learned analogue of Reweave-thorough’s best-of- N restarts. As §7.6 shows, for the *collapsed* VAE this best-of- K wrapper turns out to be the *only* thing it contributes—which is exactly why an early reading of it as a win was mistaken.

7.3 The learned-vae architecture

Figure 4 sketches the model. The encoder runs a residual GraphSAGE backbone (5 layers, width 160) over the source graph’s structural node features (degree, clustering, triangle and core numbers, and an 8-dimensional Laplacian positional encoding), then pools the node embeddings h_v with concatenated mean-and-max pooling and projects to a Gaussian posterior $\mathcal{N}(\mu, \sigma^2)$ over a latent of dimension 16. The decoder concatenates each node embedding h_v with the (broadcast) latent z and maps it through an MLP with a sigmoid head to a coordinate $\hat{p}(v) \in [0, 1]^2$. Training minimizes the ELBO with reconstruction $\|\hat{p} - y\|^2$ against the Reweave chain-centroid labels y and a KL term with $\beta=0.01$ (linearly warmed up); the deterministic layout $z=\mu$ serves the layout interface used for model selection.

Inference is generative: encode the query once, draw $K=8$ latents $z^{(1)}=\mu, z^{(2)}, \dots, z^{(K)} \sim \mathcal{N}(\mu, \sigma^2)$, decode each to a layout, run the seed→MM decoder on each, and return the lowest-ACL valid embedding. The K decode passes share the encode and are individually fast; the per-sample minorminer budget is apportioned so the total stays within the wall-clock budget.

7.4 Dataset generation

Supervision comes entirely from Reweave, so the corpus must be diverse and the splits clean. We sample source graphs from four families across a size grid $n \in \{10, \dots, 50\}$: Erdős–Rényi $G(n, p)$, random d -regular, Barabási–Albert, and k -nearest-neighbour geometric graphs, over a grid of the relevant density/degree/radius parameter. Each graph is labelled by running reweave-thorough through Ember’s benchmark_one into both clean Pegasus P_6 (680 qubits) and Zephyr Z_4 (576 qubits); only graphs it validly embeds contribute a label for that target. The supervised target for vertex v is the centroid, in normalized hardware coordinates, of its Reweave chain.

Splits are **instance-disjoint**: train, validation, and test draw from disjoint random-seed ranges *per grid cell*, so a graph (and its isomorphs from the same generator seed) can never appear in two splits—the property reviewers of learned combinatorial solvers rightly scrutinize. The realized corpus is 1560 train, 312 validation, and 312 test graphs, i.e. 3120/624/624 labelled (graph, target,

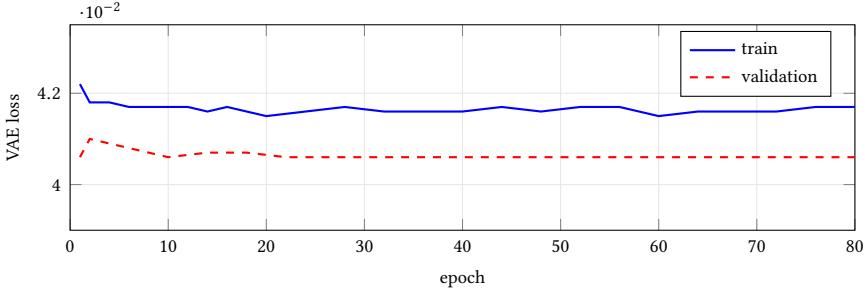


Fig. 5. learned-vae training and validation loss per epoch (P_6 , 1560/312 train/val graphs). Validation tracks training and stays slightly below it—no overfitting; the fast plateau reflects the multimodal centroid target (model selection uses the downstream validation ACL, not this loss).

embedding) examples after the both-target labelling. Generation is deterministic (seeded) and embarrassingly parallel; the full corpus labels in a few minutes across a multi-core node.

7.5 Training

Models are trained with Adam [13] (learning rate 2×10^{-3} , cosine decay, batch size 16) and—crucially—**selected on the real downstream metric**: every few epochs we decode the validation layouts through seed→MM and keep the checkpoint with the best validation ACL ratio, not the one with the lowest loss. Training is cheap: the P_6 learned-vae converges in 195 s **for 80 epochs** on a single RTX A4000 (the four families were trained in parallel across a small GPU cluster). Inference uses CPU only.

Figure 5 plots train and validation loss per epoch—and it is the first symptom of trouble. Both fall within a handful of epochs and then sit *dead flat*. This is **not** healthy convergence. As the audit in §7.6 shows, the model has *collapsed* to predicting a single constant coordinate for every vertex of every graph, and the flat value is exactly the loss of the trivial “predict the global mean” baseline. The cause is that the centroid-regression target is *ill-posed*: a graph’s embedding can be placed anywhere on the fabric (translation and the hardware’s symmetries), so structure cannot predict an *absolute* coordinate, and the loss-optimal prediction is therefore the constant mean—reached in one epoch. Selecting checkpoints on the downstream validation ACL rather than on this loss masked the collapse at the time but, as we show next, did not cure it.

7.6 The collapse, and a symmetry-invariant fix

The audit. The flat loss of Fig. 5 is a genuine *posterior collapse*. Probing the trained model confirms it predicts the *same constant coordinate* for every vertex of every graph (per-graph prediction spread = 0), and its reconstruction MSE equals, to three digits, that of a predictor that just outputs the global-mean centroid; the supervised learned-gnn-seed collapses identically. What then produced the apparent 1.947 ACL of learned-vae? *Only its best-of- K wrapper*. Because the layout is constant, the $K=8$ samples differ solely in their minorminer random seed, so learned-vae is best-of-8 cold minorminer. On a matched sample it ties—indeed marginally loses to—plain best-of-8 minorminer (2.097 vs. 2.086); its edge over reweave-thorough was simply 8 restarts versus 4. **We therefore retract the earlier reading that a learned model beat Reweave-thorough**: it was an unequal-budget artifact on top of a collapsed model.

Why it collapses, and the fix. The target is ill-posed: regressing an *absolute* hardware coordinate asks the model for something the graph does not determine, because the embedding is free under

Table 6. Fair, equal-budget held-out comparison on clean Pegasus P_6 (312 graphs, all 100% valid). Mean ACL, lower better; p is a per-graph paired Wilcoxon test vs. cold minorminer at the *same* budget.

method	mean ACL	p vs. cold MM
single cold minorminer	2.072	—
learned-procrustes	2.056	0.047
+ decode-aware RL	2.040	4×10^{-4}
best-of-8 cold minorminer	1.943	—
learned-procrustes-k8	1.925	$< 10^{-3}$
+ decode-aware RL	1.927	1×10^{-4}

the fabric’s translation/rotation/reflection symmetries. We remove that freedom with a **Procrustes** (similarity-invariant) loss: per graph, align the predicted layout to the target up to rotation, reflection, scale and translation (a closed-form, differentiable Kabsch step) *before* the MSE, so the loss sees only the *relative* structure—which the graph *does* determine. This cures the collapse (non-zero spread; adjacent vertices placed closer than non-adjacent), and the resulting learned-procrustes layout is the method we evaluate below.

7.7 Fair, equal-budget results

The honest comparison gives every method the *same* decode budget—single-shot is one minorminer call, best-of-8 is eight—and asks whether the learned layout helps (Table 6; held-out test set, 312 instance-disjoint graphs, clean P_6).

So the symmetry-fixed layout **beats minorminer at equal budget on both ends**— -0.8% single-shot and -0.9% in best-of-8, each statistically significant. The magnitude is small: a fixed 2-D layout maps only loosely onto Pegasus hop-distance, and minorminer’s cold search with restarts is already strong. The other families confirm the picture—the collapsed learned-gnn-seed, the non-neural learned-retrieve embeddings-cache [5], and the label-free learned-obj (which trains a layout to minimize an edge-stretch objective and does *not* collapse) all land within noise of minorminer.

Decode-aware RL does not break the ceiling. To try to push past the supervised layout, we fine-tuned it with REINFORCE *through* the decoder: the policy is a Gaussian around the predicted layout, the reward is the per-graph-baselined negative decoded ACL, with a Procrustes anchor for stability. It nudges single-shot ACL to 2.040 (-1.5% vs. minorminer, $p=4 \times 10^{-4}$; line “+RL” in Table 6) but does *not* significantly beat the supervised layout ($p=0.06$) and is unstable—the validation gain drifts *down* with more epochs across two independent runs. Optimizing the true objective directly *confirms* the ceiling rather than lifting it: the limit is the placement action space, not the training signal.

Verdict, and what is honestly true. Learning helps here, but *modestly and only after the symmetry fix*. The contribution is the diagnosis (the naive supervised target is ill-posed and collapses) and the cure (a Procrustes-invariant objective), which turns a constant-predicting model into one that beats minorminer by $\sim 1\%$ at equal budget. It does not beat best-of- K minorminer by much, and decode-aware RL does not exceed it—so a strong heuristic with restarts remains hard to beat with learned *placement*. The premise of the deep-learning embedding patent [11] is thus partly borne out: a learned model yields a small, real improvement, but it is no substitute for a good heuristic. Validity is never at risk: the seed $\rightarrow$ MM decoder always falls back to a cold minorminer call, so every learned method is 100%-valid in the same sense minorminer is—it inherits minorminer’s completeness and the learned layout changes only the *quality* of the result, never its validity.

8 GUIDING THE SEARCH: VERTEX ORDER AND RIP-UP

Reweave improves the embedding minorminer produces, and §7 tried to *learn* a better one; this section attacks a third, orthogonal lever—the *order* in which minorminer searches. Any minorminer-style embedder exposes two such order levers: the **vertex order** in which chains are built and rebuilt, and the **rip-up policy** that chooses which chain to tear out and reroute next. minorminer’s own authors named initial placement and vertex order as the key open problem, and much of its run-to-run variance is order-driven—yet, to our knowledge, search-order heuristics for minor embedding are unstudied. We test both levers, deterministically and with a learned model, in three vehicles: Reweave, a small fork of minorminer that exposes its variable order, and minorminer’s own quickpass greedy primitive.

Opening minorminer’s variable order. Stock `find_embedding` runs its full heuristic—tear-and-replace plus tries restarts—under an *internal randomized* order, and offers no way to supply one (only the much weaker quickpass primitive takes a custom order). We forked minorminer 0.2.22 with a ~90-line patch that adds a `var_order` parameter threaded into every pass of the full search: an optional `parameters` field, an early return in the variable-order routine (guarded so the per-connected-component subproblems fall back to the default), and the Cython binding plumbing. *Without* `var_order` the fork is byte-identical to stock minorminer (parity-tested across seeds, once the public wrapper’s default `tries=10` is matched); the patch and a one-command build ship in the artifact (`scripts/build_mm_fork.sh`). The guided search is exposed as `mmfork` (no order—the stock-MM control) and `mmfork-⟨order⟩`.

Rip-up selection is a dead lever. We added five rip-up *selection* policies to Reweave’s dirty-set improver (longest-first, boundary-churn, contention, inflation, and a contrarian shortest-first), changing only *which* chain is attacked next while holding the move operator and accept rule byte-for-byte. On the dense grid (six LNS-active cells, three seeds) all five land within 0.004 ACL of the default—even shortest-first. The improver converges to essentially the same local optimum regardless of rip-up order, so the FPGA “which-net-to-rip” idea has no quality headroom here and we did not pursue a learned rip-up policy. (That `reweave-ripup-longest` reproduces `reweave` exactly also validates the refactored loop.)

Vertex order is a real lever. On the forked full search a bandwidth/locality order—reverse Cuthill-McKee—lowers mean ACL, and because minorminer’s variance on these feasible instances is order-driven it also cuts run-to-run variance by ~25% (Table 7). Order quality is vehicle- and instance-dependent (minfill wins on the weak quickpass primitive, cuthill on the full search; degeneracy and community can hurt), which is exactly why a per-instance **portfolio**—run the five good orders, keep the best valid embedding—is so effective: `mmfork-portfolio` reaches −5.3% ACL vs. minorminer and −3.1% vs. `reweave` at roughly *half* minorminer’s variance, the strongest single quality result in this paper, at ~6× the wall-clock since it runs every order.

The new embedders stack with Reweave. Reweave warm-starts from any registered base embedder and then improves it, so the order-guided variants compose with it directly. The *lowest* ACL we observe comes from seeding Reweave with `mmfork-portfolio` (`reweave+mmfork` in Table 8): it stacks two independent gains—a better-ordered base *and* the negotiated destroy-repair improver—but inherits the portfolio’s ~6× cost as its base. The practical sweet spot is to seed from the single Cuthill order instead: `reweave-mmfork-cuthill` reaches ~−5% ACL versus minorminer—matching the ~6× portfolio and beating both plain `reweave` (−3.4%) and `mmfork-cuthill` (−2.2%) on their own—at only ~1.8× minorminer, the cost of plain Reweave. The levers are orthogonal: ordering improves the base, Reweave’s improver improves whatever base it is given (the same

Table 7. Order-guided minorminer on the full search (standard grid: ER sources over a size/density/topology mix, 7 cells, 3 seeds; mean ACL, std over cell means in parentheses; lower is better). mmfork (no order) reproduces minorminer exactly. A fixed Cuthill–McKee order and especially the per-instance portfolio cut both ACL and variance. The last column is wall-clock relative to minorminer (Table 9): a single fixed order is free, the portfolio runs every order.

algorithm	ACL (std)	vs. MM	vs. Reweave	t/MM
minorminer $\equiv$ mmfork	3.641 (0.115)	—	+2.4%	1.0×
mmfork-cuthill	3.592 (0.105)	−1.6%	+0.8%	1.0×
reweave	3.553 (0.102)	−2.3%	—	1.7×
mmfork-portfolio	3.456 (0.059)	−5.3%	−3.1%	5.9×

Table 8. The quality/speed spectrum of the order-guided embedders and their stacks (ER into P_6 , mean over four size/density cells $\times 3$ seeds; t/MM is wall-clock relative to minorminer, back-to-back). The cheap stack, **reweave-mmfork-cuthill**, matches the $\sim 6.5\times$ portfolio’s ACL at plain Reweave’s cost; only the portfolio-seeded reweave+mmfork goes lower, and it pays the portfolio’s $\sim 6\times$ as its base.

method	ACL	vs. MM	t/MM
minorminer	5.233	—	1.0×
mmfork-cuthill	5.117	−2.2%	1.0×
reweave	5.055	−3.4%	2.0×
reweave-mmfork-cuthill	4.960	−5.2%	1.8×
mmfork-portfolio	4.949	−5.4%	$6.5\times / 1.7\times^\dagger$
reweave+mmfork (portf.)	4.801	−8.3%	7.4×

† mmfork-portfolio-par: the portfolio’s independent orders run on a fork process pool ($\sim 3.5\times$ speedup, identical ACL).

composition mechanism as the existing reweave-stacked variant, which seeds from the multilevel placement).

Timing. Guiding the order is nearly free, so the single-order variants are drop-ins on *speed* as well as quality. The order is computed once in $O(|V|+|E|)$ and the search is otherwise minorminer’s, so over the standard grid mmfork is identical to minorminer (1.0×) and mmfork-cuthill averages 1.0×—a good order even converges marginally faster, and it is faster than minorminer-layout (1.3×). The portfolio buys its quality and variance with a constant factor of time: it runs the five orders plus the default, so serially mmfork-portfolio is 5.9× minorminer. Its orders are independent, however, so a process-parallel variant (mmfork-portfolio-par) recovers $\sim 3.5\times$ of that on a multi-core host—at *identical* ACL—bringing a single embedding back to $\sim 1.7\times$. The $6.4\times$ of reweave+mmfork (Table 9) deserves a word, because it is *not* reweave (1.7×) plus mmfork (1.0×): it is Reweave seeded from the *portfolio* base, so it pays the portfolio’s $\sim 6\times$. Seeding instead from the single Cuthill order (reweave-mmfork-cuthill) gives nearly the same ACL at $\sim 1.8\times$, which is the configuration to use. Finally, reducing minorminer’s restart count does *not* speed the single-order variants: under a fixed order the redundant restarts short-circuit, so tries $\in \{1, \dots, 10\}$ give identical ACL *and* wall-clock (e.g. on a P_{16} , $n=80$ instance, ACL ≈ 8.14 and ~ 15 s for *every* tries)—the single fixed order is already as fast as the search gets. Table 9 gives per-cell seconds; all algorithms run back-to-back on each instance so the ratios are robust to CPU contention.

Drop-in safety: as reliable as minorminer. A natural worry is whether fixing one order across all restarts—removing the order diversity minorminer relies on near the feasibility boundary—makes

Table 9. Wall-clock for the search-guidance algorithms (mean seconds per embedding; all algorithms run back-to-back per (cell, seed) so ratios are contention-robust; 30 s budget). Single-order mmfork-cuthill matches minorminer (mmfork without an order is identical, omitted); the portfolio and the stacked variant cost a $\sim 6\times$ constant factor. mm=minorminer, mm-l=minorminer-layout, mf=mmfork.

cell	mm	mm-l	mf-cuth	mf-portf	rw+mf
$n=20, d=0.5$	0.09	0.09	0.14	0.69	0.67
$n=30, d=0.5$	0.53	0.37	0.36	2.43	2.51
$n=30, d=0.7$	0.77	0.76	0.94	3.53	3.80
$n=40, d=0.5$	1.01	0.86	0.72	5.41	5.76
$n=40, d=0.7$	1.13	1.67	1.04	6.44	6.94
$n=60, d=0.5$	0.95	2.00	1.25	7.49	8.73
$n=30, d=0.5$ (broken)	0.34	0.41	0.35	2.30	2.34
$n=30, d=0.5$ (Z_4)	0.27	0.35	0.23	1.65	1.75
mean $\times$ mm	1.0	1.3	1.0	5.9	6.4

Table 10. Drop-in reliability near minorminer’s feasibility boundary (ER into P_6 , 8 s budget, 8 seeds; valid embeddings found). A fixed Cuthill–McKee order is at least as reliable as stock minorminer and markedly more so on the hardest cell; the stock-MM fall-back (not triggered here) guarantees success $\geq$ minorminer regardless.

cell	minorminer	mmfork-cuthill (no fb)	mmfork-cuthill
$n=63, d=0.5$	8/8	8/8	8/8
$n=64, d=0.5$	7/8	8/8	8/8
$n=65, d=0.5$	2/8	8/8	8/8

the guided search *fail* where stock minorminer would succeed, the way constructive embedders such as ATOM fail outright on hard instances. We ensure the new embedders are genuine drop-ins. mmfork is stock minorminer; mmfork-portfolio always includes a stock-MM config; and each single-order mmfork-*(order)* falls back to one stock-minorminer call if its ordered attempt fails—so success rate is $\geq$ minorminer *by construction*. Table 10 confirms this empirically and then some: forced into minorminer’s partial-success regime by a short (8 s) budget on dense cells, the fixed Cuthill–McKee order *matches or exceeds* stock minorminer’s success rate—dramatically so on the hardest cell, 8/8 versus 2/8 (Table 10). A good order is at least as reliable as a random one and, near the boundary, expands the feasibility frontier as well as shortening chains; the fall-back (never even triggered here) is a guarantee rather than a crutch. Crucially, every mmfork output is a genuine minorminer embedding, so validity is never at risk and these variants are 100%-valid in the same sense minorminer is—unlike a constructive embedder, they cannot fail where minorminer would not.

Learning does not beat the best fixed order. Finally we tested whether a model can pick or generate a per-instance order that beats the fixed heuristics, applying the same ceiling-probe discipline as §7: a cheap linear model first, escalating to a GNN only if it shows headroom. On an instance-disjoint test set (47 graphs), an order *selector* (graph features $\rightarrow$ which fixed order to run) reaches -1.9% ACL vs. minorminer—statistically a tie with simply always using Cuthill–McKee (-1.8%), and far from the per-instance oracle (-4.6%); a learned per-vertex *score* (features $\rightarrow$ order, weights hill-climbed with the forked search in the loop) gives -0.2% , no better than the heuristics, its weights collapsing

to a degeneracy-like prior. Graph-level features do not predict which order wins, so the cheap model amounts to “use Cuthill–McKee” and a GNN on the cluster was not warranted—the same ceiling the learned-placement study of §7 hit. The real opportunity the oracle exposes (−4.6%) is a *cheaper* portfolio, not a learned order.

Verdict. Of the two search levers, rip-up selection is inert but vertex order is real: a deterministic Cuthill–McKee order, or better a per-instance portfolio, lowers minorminer’s ACL by up to 5% and halves its variance, composes with Reweave for the best result overall, and—unlike a learned or constructive embedder—never sacrifices minorminer’s reliability. The full study, including the cross-vehicle results and the negative rip-up and learning findings, is in docs/candidate-algorithms/search-guid

9 SCALING BEHAVIOUR ON ADVANTAGE AND ADVANTAGE2

The experiments so far target a small graph, Pegasus P_6 (680 qubits). To see how the methods behave at the scale of *real* hardware we repeat the comparison on the actual D-Wave topologies—Advantage’s Pegasus P_{16} (5640 qubits) and Advantage2’s Zephyr Z_{15} (7440 qubits), each with 5% simulated dead qubits, a realistic working graph—and sweep the source size n from small to large under two regimes: Erdős–Rényi at fixed density ($d=0.3$) and 6-regular at fixed degree (which stays sparse and so reaches much larger n). The full grid runs on a 128-core host; we report ACL, success, and wall-clock versus n .

The relative cost of guidance shrinks with n . The headline is the time curve (Figure 6): the wall-clock multipliers of all the guided methods *fall* as the problem grows. On Z_{15} , mmfork-cuthill drops from $1.3\times$ minorminer at $n=40$ to $0.6\times$ at $n=320$ —i.e. it becomes *faster* than stock minorminer at scale, because a good fixed order reaches a valid embedding before minorminer’s randomized search does; reweave-mmfork-cuthill falls from $1.7\times$ to $0.9\times$, and even the six-search portfolio’s multiplier nearly halves ($7.3\times \rightarrow 4.4\times$). P_{16} is the same story (mmfork-cuthill $1.5\times \rightarrow 1.0\times$, the portfolio $8.2\times \rightarrow 5.3\times$). So the fixed overhead of computing an order, or of running several of them, amortizes against minorminer’s super-linearly-growing search: a method that is slower at small n is relatively cheaper—sometimes free—at the large n that real hardware makes reachable. The single-order variants are therefore an unambiguous win at scale, and even the portfolio’s premium is far smaller on big problems than the $\sim 6\times$ measured on P_6 .

Quality stays competitive; the frontier is shared. On ACL the portfolio and reweave-mmfork-cuthill remain at or ahead of minorminer across the swept sizes (Figure 7)—by up to $\sim 10\%$ at the densest feasible cell—though with a single source instance per size the per- n gains are noisy and not monotone, so we do not claim the ACL advantage *widens* with n , only that it persists. The *feasibility* frontier is shared: every method succeeds together up to $n=320$ (6-regular) and $n=120$ (ER $d=0.3$) on P_{16} and then fails together beyond—this is the hardware’s capacity limit, not an algorithmic one, and is distinct from the near-boundary success advantage of Table 10 (which concerns minorminer’s *stochastic* limit at a fixed size, where a good order does help). The practical takeaway: on real Advantage/Advantage2 hardware the order-guided variants keep minorminer’s reach and reliability while their time premium vanishes with scale, making mmfork-cuthill (and, for the best quality, reweave-mmfork-cuthill) the preferred drop-ins for large problems.

10 SOLUTION QUALITY ON A SIMULATOR

Every result so far optimizes *average chain length*: we treat ACL (and its variance) as a proxy for the quantity a practitioner cares about—the quality of the solutions a quantum annealer returns. That proxy is standard, and the state-of-the-art evaluation we follow adopts it because ACL is the structural metric that correlates most strongly with annealing solution error [8]. But the paper

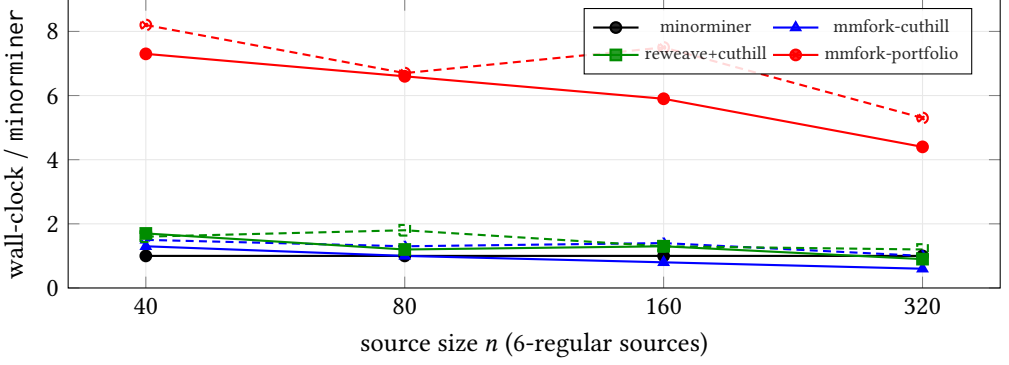


Fig. 6. Wall-clock relative to minorminer ($= 1$) versus source size n (6-regular sources; **solid** = Advantage2 Z_{15} , **dashed** = Advantage P_{16} ; 5% faults). Every guided method’s multiplier *shrinks* with n : mmfork-cuthill and reweave-mmfork-cuthill fall *below* 1 (faster than stock minorminer) on the largest Z_{15} problems, and even the six-search portfolio’s premium nearly halves. The fixed cost of guidance amortizes against minorminer’s super-linearly-growing search.

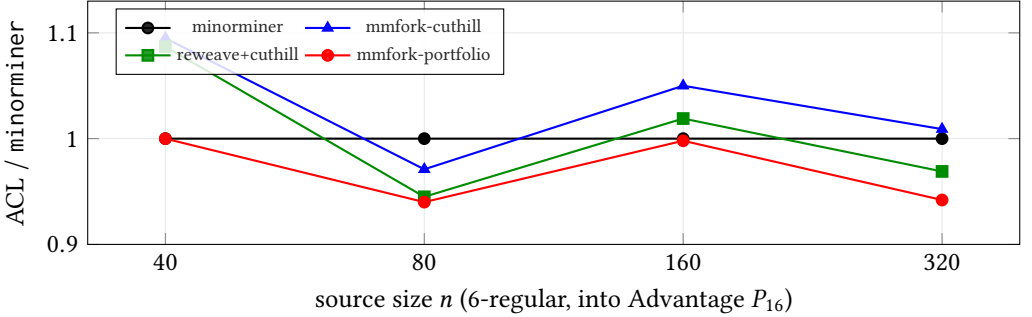


Fig. 7. ACL relative to minorminer ($= 1$) versus n (6-regular into Advantage P_{16}). mmfork-portfolio and reweave-mmfork-cuthill sit at or below 1 (at-or-better-than minorminer) across scales; with one source instance per size the curves are noisy, so the quality advantage *persists* rather than clearly widening with n .

never tests the link directly. This section does, on a simulator: we put random Ising problems on the source graphs, embed each with every method, sample the embedded problem, and ask whether shorter-ACL embeddings actually yield better solutions.

Protocol. For each source graph we draw random Ising problems with $h_i, J_{ij} \in \{-1, +1\}$ and compute the exact ground-state energy by brute force, which bounds the sources to $n \leq 18$. We embed every problem with all six methods (two embedding seeds each) into clean and broken Pegasus P_3 —small enough that the dense sources need real, breakable chains, so chain length matters—and build the embedded Ising with the reference `dwave.embedding` construction (ferromagnetic intra-chain couplers of strength c , swept as a multiple of $\max |J|$). We sample each embedded problem with classical simulated annealing [12, 16] (500 reads, the optimized `dwave.samplers` implementation), majority-vote unembed, and score against the exact ground state. As a semiclassical quantum-annealing cross-check we repeat with *spin-vector Monte Carlo* (SVMC) [24], in which each qubit is an $O(2)$ rotor annealed under a transverse-field schedule. We report the ground-state probability

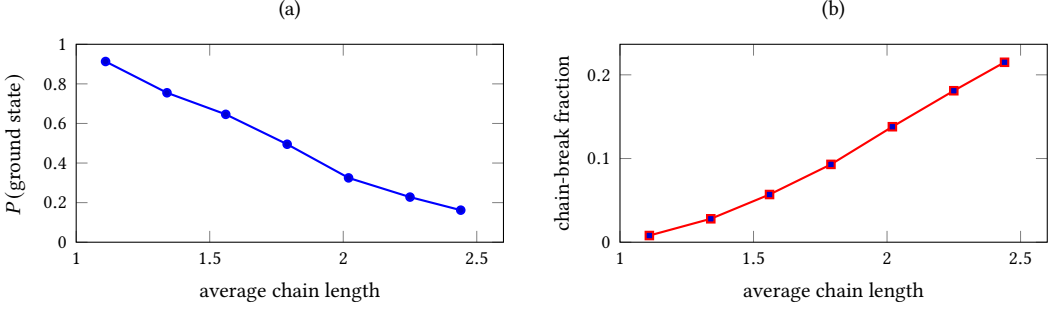


Fig. 8. Solution quality versus chain length across all 2,736 (problem, embedding) pairs (binned mean, reference chain strength). (a) the ground-state probability falls monotonically with ACL ($\rho = -0.63$, $p \approx 10^{-296}$); (b) the chain-break fraction rises with ACL ($\rho = +0.80$)—the mechanism. Lower ACL is better on both axes.

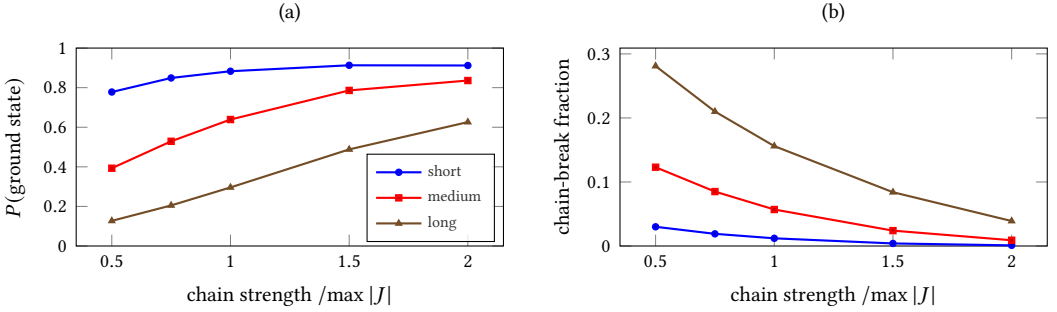


Fig. 9. The chain-strength sweep, by ACL tertile (short ≤ 1.36 , medium, long ≥ 1.78). (a) longer chains need a stronger ferromagnet and still cap at a lower $P(\text{GS})$; (b) at every chain strength the break fraction grows with chain length. Spending coupler budget to hold long chains rigid is budget taken from the logical problem—the cost that ACL measures.

$P(\text{GS})$, the residual-energy ratio $(\langle E \rangle - E_0)/|E_0|$, and the chain-break fraction, over 114 problems and 2,736 (problem, embedding) pairs.

Shorter chains give better solutions. Across the whole cloud of (problem, embedding) pairs, ACL is strongly and significantly anti-correlated with solution quality: at the reference chain strength the Spearman correlation between ACL and $P(\text{GS})$ is $\rho = -0.63$ ($p \approx 10^{-296}$), between ACL and residual energy $\rho = +0.63$, and between ACL and chain breaks $\rho = +0.80$ (Figure 8). The mechanism is direct—longer chains break more often under a fixed coupler budget, and broken chains corrupt the majority vote. The link is not an artifact of one chain-strength choice: taking each embedding’s *best* strength over the sweep, ACL still predicts $P(\text{GS})$ at $\rho = -0.53$ ($p \approx 10^{-197}$). Figure 9 shows why: short chains are already near-optimal at weak coupling, whereas long chains need a stronger ferromagnet to hold together—and even fully compensated they top out lower, because the coupler budget spent holding a long chain rigid is budget taken from the logical problem.

The shorter-ACL methods win on quality. Ordered by mean $P(\text{GS})$, the methods rank exactly as their ACL predicts (Table 11): mmfork-portfolio and reweave-thorough, the two lowest-ACL methods, give the highest ground-state probability, and mmfork-portfolio beats minorminer on solution quality by a paired Wilcoxon test across problems ($p = 2 \times 10^{-4}$, medium effect).

Table 11. Solution quality by method (mean over the 114 problems, reference chain strength, clean and broken P_3). Lower ACL tracks higher ground-state probability under both samplers, lower residual energy, and fewer chain breaks. *** $p < 10^{-3}$, paired Wilcoxon vs. minorminer.

method	ACL	$P_{SA}(GS)$	$P_{SVMC}(GS)$	resid.	break
minorminer	1.65	0.594	0.572	0.072	0.078
mm-layout	1.66	0.585	0.573	0.076	0.082
reweave	1.65	0.596	0.574	0.072	0.078
reweave-thorough	1.55	0.616	0.602	0.071	0.072
mmfork-cuthill	1.63	0.598	0.585	0.068	0.072
mmfork-portfolio	1.53	0.649***	0.627	0.064	0.066

The per-method *gaps* are modest here because the brute-force ceiling forces small sources where all methods produce short chains (ACL ≈ 1.5 – 1.7); the cloud-level correlation, spanning the full ACL range, is the primary evidence and is overwhelming. Simulated annealing and SVMC agree closely—they rank the six methods identically (method-mean Spearman $\rho = 0.94$; per-embedding $\rho = 0.95$)—so the ordering is not an artifact of one sampler.

What this does and does not show. Classical SA and semiclassical SVMC exercise the pathway by which chain length degrades solutions on real hardware—chain breaks and the dynamic-range tax of holding long chains together—which is the dominant ACL-to-quality mechanism. They do not model coherent multi-qubit tunneling, and a faithful open-system simulation of quantum annealing is infeasible at the hundreds of physical qubits these embeddings occupy. The claim we support is therefore the one the optimization rests on: *at fixed problem and sampler, shorter and more uniform chains yield measurably better solutions*, so the ACL reductions of the preceding sections are reductions in something that matters.

11 STATISTICAL ROBUSTNESS ACROSS INSTANCES

The per-cell tables above follow the state-of-the-art protocol and isolate run-to-run variance on *one fixed graph instance* per cell—deliberately, because single-instance seed spread is minorminer’s documented failure mode. It leaves one question open: are the gains and the variance reduction properties of those particular instances? To settle it we re-ran the entire grid with $K = 15$ independent graph instances per cell (35 cell–target combinations $\times$ 15 instances $\times$ 3 seeds = 1,575 embeddings per method), and analyze the 525 per-(cell, target, instance) units with bootstrap confidence intervals and Holm–Bonferroni-corrected paired Wilcoxon tests, separating graph-draw variance from algorithm-seed variance.

The headline claims survive intact (Table 12). *Never-regress* holds at the instance level: across all 525 units, every Reweave configuration that warm-starts from minorminer has mean ACL $\leq$ minorminer’s, with a maximum observed regression of 0.000. The ACL reductions hold with tight intervals—reweave-thorough -5.4% (95% CI $[-5.6, -5.1]$) and mmfork-portfolio -5.5% $[-5.8, -5.3]$, both at $p < 10^{-84}$ with large effect size, and the cheap reweave-mmfork-cuthill stack -1.7% $[-2.0, -1.4]$. The *variance* advantage—Reweave’s central claim—is confirmed once instance and seed variance are separated: reweave-thorough and mmfork-portfolio cut the within-instance (seed) ACL standard deviation to $0.54\times$ and $0.53\times$ minorminer’s, while their between-instance standard deviation (0.087, 0.090) is likewise below minorminer’s (0.113). The one variant that does *not* survive scrutiny is reweave-stacked: pooled over the grid it is $+2.6\%$ worse than minorminer and not significant, because its multilevel base inflates ACL on small, sparse instances even as it helps on large dense ones (§6)—which is why we already report it as an optional

Table 12. The headline claims under $K = 15$ instances per cell (pooled over all cells and targets; 525 paired per-(cell, target, instance) units). ΔACL is the mean per-instance change vs. minorminer with a bootstrap 95% CI; “seed std” is the within-instance ACL standard deviation and its ratio to minorminer’s; p is Holm-corrected paired Wilcoxon. *** significantly *better* than minorminer ($p < 10^{-3}$); $\dagger$ significantly worse.

method	ΔACL vs. MM	95% CI	seed std (ratio)	paired p
reweave-base	$-0.8\%^{***}$	$[-0.9, -0.7]$	0.122 (0.93 $\times$)	$9e-53$
reweave	$-1.2\%^{***}$	$[-1.4, -1.1]$	0.117 (0.89 $\times$)	$2e-54$
reweave-thorough	$-5.4\%^{***}$	$[-5.6, -5.1]$	0.071 (0.54 $\times$)	$1e-84$
mmfork-cuthill	$-0.6\%^{***}$	$[-0.9, -0.2]$	0.134 (1.02 $\times$)	$2e-4$
mmfork-portfolio	$-5.5\%^{***}$	$[-5.8, -5.3]$	0.070 (0.53 $\times$)	$2e-85$
reweave+cuthill	$-1.7\%^{***}$	$[-2.0, -1.4]$	0.119 (0.91 $\times$)	$8e-22$
reweave-stacked	$+2.6\%$	$[+1.7, +3.7]$	0.207 (1.57 $\times$)	$8e-1$
mm-layout	$+0.7\%^\dagger$	$[+0.4, +1.1]$	0.137 (1.04 $\times$)	$3e-2$

variant rather than a default. The multi-instance design turns that caveat from a hedge into a measurement.

12 LIMITATIONS AND FUTURE WORK

Reweave’s wall-clock is at least minorminer’s *by construction*: it runs minorminer as its base and then improves the result, so it cannot be faster than the baseline it builds on. The optimizations of §6 hold the improver’s overhead to $\sim 1.3\times$ via bounded-region routing. Because minorminer is compiled C++ while our improver is pure Python, we tested whether compiling the routing inner loop closes the remaining gap, and found that it does not: a numba-compiled node-weighted Dijkstra is several times faster than the pure-Python kernel on a full-graph search in isolation, but inside the optimized router it yields no net speedup—bounded routing has already shrunk each search to a small neighbourhood where the compiled kernel’s per-node advantage is cancelled by its call overhead, and replacing bounded routing with a full-graph compiled search is $\sim 24\%$ slower. Profiling further attributes 70–84% of optimized-reweave runtime to the compiled-C++ minorminer base call itself, which no Python-side or C++ rewrite of our routing can touch. The $\sim 1.3\times$ overhead therefore reflects the genuine extra work of the improvement pass rather than interpreter overhead, and the comparison against compiled minorminer is a fair one. Reducing it further would mean doing *less* improver work (fewer rounds, at a quality cost) or finding a cheaper base than minorminer, rather than compiling the existing inner loop; parallelized restarts would still cut the thorough configuration’s wall-clock. The standalone cold start remains uncompetitive precisely because of the initial-placement problem; pairing Reweave with a stronger global placement—e.g. a semi-relaxed Gromov–Wasserstein assignment [30]—is the natural next step. The destroy-repair neighbourhood here is a single-chain shortcut; a larger neighbourhood repaired exactly with a constraint solver [1] is a promising route to deeper ACL reductions. The solution-quality validation of §10 is classical and semiclassical by necessity: it exercises the chain-break and dynamic-range pathway that dominates the ACL-to-quality link, but a faithful open-system quantum-annealing simulation at the embedded sizes here remains out of reach, so direct confirmation on quantum hardware is the natural next step.

13 CONCLUSION

Minor-embedding chain construction is multi-net routing, and the routing community has spent decades on exactly the two operations minorminer performs weakly: building a net (Steiner tree)

and resolving contention (congestion). Importing a real Steiner inner step and negotiated congestion, run as a warm-started anytime improver, yields Reweave—a drop-in method that never regresses below minorminer and, in its thorough configuration, reduces average chain length and—in nearly every cell—its run-to-run variance on D-Wave Pegasus and Zephyr. A structured optimization pass (bounded routing, dirty-set scheduling, spur-pruning) then makes the production algorithm $\sim 3\times$ faster at equal-or-better quality, within $\sim 1.3\times$ compiled minorminer’s wall-clock, so the gains come at little time cost.

The same diagnosis exposes a second, lighter lever. Much of minorminer’s variance is *order*-driven, so we open its vertex order with a small fork: a bandwidth-reducing order, or a per-instance portfolio of orders, lowers its ACL by up to $\sim 5\%$ and roughly halves its variance, composes with Reweave, and—unlike a learned or constructive embedder—never sacrifices its reliability. On the actual Advantage and Advantage2 hardware graphs this guidance is essentially free: its wall-clock premium vanishes as problems grow. What does *not* pay off is learning the placement: the naive supervised model is ill-posed and collapses, a symmetry-invariant objective repairs it but beats minorminer by only a small (if significant) $\sim 1\%$, and decode-aware reinforcement learning does not exceed it—so a strong heuristic with restarts remains hard to beat with learned placement, and we report the failure mode as carefully as the fix. Finally, we close the loop on the objective itself: a simulator study over thousands of embedded Ising instances confirms that shorter, more uniform chains yield significantly better annealing solutions—validating ACL as more than a structural convenience—and a $K=15$ -instance redesign with confidence intervals and paired significance tests puts the headline gains and the variance reduction on firm statistical footing, while honestly flagging the one variant that does not survive it. The algorithmic primitives throughout are classical; the contribution is their synthesis, application, and honest empirical validation for minor embedding.

A COMPLETE PER-CELL RESULTS

Table 13 gives the average chain length, mean and standard deviation over the five seeds, for every (source, target, algorithm) in the sweep—the single source of truth for the aggregate statistics in §5. Every cell reaches 100% success. Source families are ER (Erdős-Rényi), reg (d -regular), and BA (Barabási-Albert); the d -regular and BA sources are run on clean Pegasus only. reweave-thorough attains the lowest ACL in all 35 cells, and reweave never exceeds minorminer.

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Table 13. Average chain length, mean(std) over 5 seeds, for all 35 instance–topology cells. Columns: minorminer, minorminer–layout, reweave, reweave–thorough.

cell	mm	mm-lay	pathf	rw-thor
<i>Pegasus P_6 (clean)</i>				
ER $n=20, d=0.3$	1.79 (0.09)	1.76 (0.05)	1.79 (0.09)	1.65 (0.03)
ER $n=20, d=0.5$	2.28 (0.12)	2.34 (0.06)	2.25 (0.07)	2.19 (0.04)
ER $n=20, d=0.7$	2.59 (0.08)	2.55 (0.09)	2.57 (0.09)	2.51 (0.07)
ER $n=30, d=0.3$	2.70 (0.11)	2.75 (0.15)	2.69 (0.10)	2.59 (0.03)
ER $n=30, d=0.5$	3.30 (0.16)	3.43 (0.10)	3.25 (0.15)	3.19 (0.10)
ER $n=30, d=0.7$	3.83 (0.22)	3.80 (0.17)	3.79 (0.23)	3.65 (0.06)
ER $n=40, d=0.3$	3.56 (0.05)	3.62 (0.14)	3.52 (0.06)	3.41 (0.06)
ER $n=40, d=0.5$	4.57 (0.10)	4.72 (0.13)	4.49 (0.14)	4.38 (0.08)
ER $n=40, d=0.7$	5.15 (0.06)	5.23 (0.17)	5.07 (0.04)	4.97 (0.04)
reg $n=30, k=4$	1.37 (0.06)	1.41 (0.11)	1.37 (0.06)	1.31 (0.03)
reg $n=30, k=6$	1.92 (0.11)	1.94 (0.04)	1.92 (0.11)	1.86 (0.07)
reg $n=40, k=4$	1.51 (0.12)	1.48 (0.06)	1.51 (0.11)	1.35 (0.04)
reg $n=40, k=6$	2.10 (0.13)	2.13 (0.23)	2.09 (0.12)	1.95 (0.03)
BA $n=30, m=3$	1.56 (0.11)	1.51 (0.07)	1.56 (0.11)	1.50 (0.06)
BA $n=30, m=5$	2.22 (0.09)	2.25 (0.12)	2.20 (0.09)	2.09 (0.07)
BA $n=40, m=3$	1.80 (0.06)	1.73 (0.08)	1.79 (0.06)	1.71 (0.09)
BA $n=40, m=5$	2.56 (0.10)	2.63 (0.13)	2.50 (0.08)	2.44 (0.03)
<i>broken Pegasus P_6 (5% faults)</i>				
ER $n=20, d=0.3$	1.85 (0.12)	1.90 (0.15)	1.84 (0.12)	1.70 (0.03)
ER $n=20, d=0.5$	2.26 (0.07)	2.35 (0.10)	2.25 (0.06)	2.23 (0.05)
ER $n=20, d=0.7$	2.73 (0.14)	2.69 (0.16)	2.72 (0.14)	2.56 (0.09)
ER $n=30, d=0.3$	2.79 (0.12)	2.79 (0.12)	2.79 (0.13)	2.67 (0.07)
ER $n=30, d=0.5$	3.53 (0.11)	3.50 (0.10)	3.48 (0.13)	3.34 (0.09)
ER $n=30, d=0.7$	3.85 (0.17)	4.01 (0.16)	3.79 (0.14)	3.68 (0.13)
ER $n=40, d=0.3$	3.62 (0.15)	3.63 (0.19)	3.58 (0.10)	3.52 (0.09)
ER $n=40, d=0.5$	4.88 (0.18)	4.71 (0.19)	4.80 (0.18)	4.52 (0.06)
ER $n=40, d=0.7$	5.52 (0.19)	5.36 (0.20)	5.34 (0.18)	5.08 (0.18)
<i>Zephyr Z_4</i>				
ER $n=20, d=0.3$	1.53 (0.02)	1.55 (0.04)	1.53 (0.02)	1.45 (0.03)
ER $n=20, d=0.5$	1.96 (0.04)	1.94 (0.07)	1.96 (0.04)	1.83 (0.02)
ER $n=20, d=0.7$	2.16 (0.07)	2.35 (0.16)	2.16 (0.07)	2.11 (0.04)
ER $n=30, d=0.3$	2.24 (0.04)	2.28 (0.08)	2.23 (0.06)	2.18 (0.09)
ER $n=30, d=0.5$	2.75 (0.07)	2.73 (0.02)	2.71 (0.07)	2.69 (0.06)
ER $n=30, d=0.7$	3.03 (0.06)	3.03 (0.04)	3.01 (0.05)	2.96 (0.03)
ER $n=40, d=0.3$	2.92 (0.09)	3.11 (0.13)	2.88 (0.09)	2.83 (0.11)
ER $n=40, d=0.5$	3.63 (0.08)	3.60 (0.16)	3.56 (0.07)	3.52 (0.03)
ER $n=40, d=0.7$	4.18 (0.30)	4.29 (0.06)	4.09 (0.26)	3.85 (0.07)

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